

Temporal Point Processes and Races of Clocks

Hazards, survival, likelihoods with silence, and why a race is a posterior over causes

Why this tutorial

Our models predict *when* the next event happens and *what* it is. The mathematical language for that is the temporal point process. A *race of clocks*, several candidate events competing to happen first, is the core primitive of our model family. This tutorial builds the vocabulary from scratch: hazards, survival, the likelihood of an event sequence (including the information carried by silence), marks, and races. It ends with the single most important identity for learning: the gradient of a race’s log-likelihood is “responsibility minus exposure”.

1 Events in continuous time

A *temporal point process* is a random sequence of event times $0 < t_1 < t_2 < \dots$. Its counting process is $N(t) = \#\{k : t_k \leq t\}$. Write $\mathcal{H}_t$ for the history up to (but not including) t .

Definition 1 (Conditional intensity / hazard). The *intensity* $\lambda(t)$ is the instantaneous event rate given the history:

$$\lambda(t) dt = \Pr\{\text{an event in } [t, t + dt) \mid \mathcal{H}_t\}.$$

For the waiting time τ since the last event, the same object is called the *hazard* $h(\tau)$.

The hazard determines everything else about the next waiting time:

$$\begin{aligned} \text{integrated hazard: } H(\tau) &= \int_0^\tau h(s) ds, \\ \text{survival (no event yet): } S(\tau) &= \Pr\{\tau_{\text{next}} > \tau\} = e^{-H(\tau)}, \\ \text{density of the waiting time: } f(\tau) &= h(\tau) S(\tau). \end{aligned}$$

Derivation. $S(\tau + d\tau) = S(\tau) (1 - h(\tau) d\tau)$, so $S' = -hS$ and $S = e^{-H}$; then $f = -S' = hS$.

Example 2 (Exponential clock). Constant hazard $h = \lambda$: $H = \lambda\tau$, $S = e^{-\lambda\tau}$, $f = \lambda e^{-\lambda\tau}$, mean $1/\lambda$. The exponential is *memoryless*: $\Pr\{\tau > a + b \mid \tau > a\} = e^{-\lambda b}$.

Example 3 (Shaped clocks). Weibull: $h(\tau) = \lambda k \tau^{k-1}$ (increasing hazard if $k > 1$: “due” events). Log-normal: $\log \tau \sim \mathcal{N}(\mu, \sigma^2)$, a *delayed* clock that rarely fires early. Our models use mixtures of exponential and delayed (log-normal) clocks.

2 The likelihood of a sequence, and the information in silence

Observe events at $t_1, \dots, t_n$ in a window $[0, T]$. The probability density of exactly this observation is

$$L = \left(\prod_{k=1}^n \lambda(t_k) \right) \exp\left(- \int_0^T \lambda(t) dt \right), \quad \log L = \sum_k \log \lambda(t_k) - \int_0^T \lambda(t) dt.$$

The integral term is the log-probability of *no events* in all the gaps, including the final stretch $(t_n, T]$.

Key idea

Silence is data. The integral $\int \lambda$ penalises a model that expects events where none occurred. A model that ignores silence can assign high likelihood to every observed event while predicting far too many unobserved ones. We call training that keeps this term *silence-aware supervision*.

Marks. If each event carries a type (mark) $k \in \{1, \dots, K\}$, give each mark its own intensity $\lambda_k(t)$. The marked log-likelihood is $\sum_{\text{events}} \log \lambda_{k_j}(t_j) - \sum_k \int_0^T \lambda_k$. Equivalently, the total rate $\Lambda = \sum_k \lambda_k$ generates times and the mark is drawn with probability λ_k/Λ .

Recording resolution. Real timestamps are rounded (to seconds, days). A density model can put enormous density on the rounding lattice and gain likelihood without predicting anything. Comparing models fairly requires scoring at a stated resolution or *dequantizing* (spreading each recorded time uniformly within its recording cell). This matters in practice: on some public benchmarks a large part of published likelihood comes from the lattice.

3 Races of clocks

Definition 4 (Race). K clocks run in parallel from the same start. Clock i would fire after an independent waiting time with hazard h_i . The race outputs the *first* firing time $\tau = \min_i \tau_i$ and the *winner* $I = \arg \min_i \tau_i$.

Proposition 5 (Race laws). With $H_i(\tau) = \int_0^\tau h_i$:

1. The race's survival is the product $S(\tau) = \prod_i S_i(\tau) = e^{-\sum_i H_i(\tau)}$; its hazard is the sum $\sum_i h_i(\tau)$.
2. The joint density of (time, winner) is $f(\tau, i) = h_i(\tau) S(\tau)$.
3. Given that the first event happens at τ , the winner is clock i with probability

$$\rho_i(\tau) = \frac{h_i(\tau)}{\sum_j h_j(\tau)}.$$

Proof. (1) Independence: no clock has fired by τ iff each has not. (2) Clock i fires in $[\tau, \tau + d\tau]$ with probability $h_i(\tau) S_i(\tau) d\tau$ and the others have not fired: multiply by $\prod_{j \neq i} S_j$. (3) Divide (2) by its sum over i . $\square$

Example 6 (Exponential race = softmax). With constant rates $\lambda_i = e^{\theta_i}$, $\rho_i = e^{\theta_i} / \sum_j e^{\theta_j}$ does not depend on τ , so the winner is a softmax draw, independent of the time, and $\tau \sim \text{Exp}(\sum \lambda_j)$. This is the ‘‘Gumbel-max’’ fact in disguise: $-\log \tau_i = \theta_i + \text{Gumbel noise}$, and the arg-max of noisy scores is softmax-distributed.

4 Learning: responsibility minus exposure

Parametrize each clock's scale: $h_i(\tau) = e^{\theta_i} k_i(\tau)$, so $H_i = e^{\theta_i} K_i$ with $K_i = \int_0^\tau k_i$. For one observed first event at τ (winner unobserved, or marginalized), $\log f(\tau) = \log \sum_j h_j(\tau) - \sum_j H_j(\tau)$. Then

$$\frac{\partial \log f(\tau)}{\partial \theta_i} = \rho_i(\tau) - H_i(\tau)$$

Responsibility ρ_i (the posterior probability that clock i caused the event) minus *exposure* H_i (how much clock i was at risk without firing). If the winner is observed (a marked event), ρ_i is replaced by the indicator $1[I = i]$.

Key idea

The gradient of a race is an inference result: it attributes the observed event to its possible causes. That is literally credit assignment, and at the output of a race it is exact and local: each clock needs only its own hazard and the race's total rate. Note that the expected gradient is zero: $\mathbb{E}[\rho_i] = \mathbb{E}[H_i]$ (both equal the probability that i wins), the usual zero-mean property of a score.

Exercises

Exercise 1. Show that the minimum of independent exponentials with rates λ_i is exponential with rate $\sum \lambda_i$.

Solution. $\Pr\{\min > \tau\} = \prod e^{-\lambda_i \tau} = e^{-(\sum \lambda_i) \tau}$.

Exercise 2. For two Weibull clocks with the same shape k , show the winner is independent of τ . Is this true for different shapes?

Solution. $\rho_i = \lambda_i k \tau^{k-1} / \sum_j \lambda_j k \tau^{k-1} = \lambda_i / \sum \lambda_j$. With different shapes ρ_i depends on τ : early events favour the clock with the decreasing hazard.

Exercise 3. Write the log-likelihood of a sequence observed on $[0, T]$ with no events at all. What does maximizing it do?

Solution. $\log L = -\int_0^T \lambda$; it drives the predicted rate toward zero: silence teaches the model that events are rare.

Exercise 4. Derive $\partial \log f / \partial \theta_i = \rho_i - H_i$ for the race above.

Solution. $\partial_{\theta_i} h_j = \delta_{ij} h_i$ and $\partial_{\theta_i} H_j = \delta_{ij} H_i$, so $\partial \log \sum h = h_i / \sum h = \rho_i$ and $\partial(-\sum H) = -H_i$.

Where this appears in our work

Race-of-clocks heads in `experiments/tpp/race_tpp*.py` (EasyTPP battle B1, neural-TPP benchmark B4); silence-aware supervision throughout; the recording-resolution floor of our clocks and the B4 lattice audit; theory note 160 §2 (race credit is posterior inference); note 168 (Fisher geometry of races, Tutorial 6).